TRAN TRUNG KHANG

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EDUCATION

NUS – National University of Singapore, Singapore — Aug 2025 – Aug 2029
Bachelor of Computing (Honours) in Artificial Intelligence
Double Major in Computer Science and Mathematics
Special Programme in Mathematics (SPM) — **CGPA**: 4.92/5.0

VNUHCM-High School for the Gifted, Ho Chi Minh City, Vietnam — Aug 2021 – Jun 2024
Student in Specialized Math Class — **GPA**: 10.0/10.0

EXPERIENCE

Mathematical AI Group @ NUS | Singapore — Apr 2025 – Present
Research Intern | Supervisor: Prof. Tan NGUYEN (tanmn@nus.edu.sg)
- Research topics: Uncertainty Quantification, Conformal Prediction, Tree-sliced Wasserstein.

NUS Student Work Scheme (NSWS) | Singapore — Jan 2026 – Dec 2026
Research Assistant | Supervisor: Prof. Jonathan SCARLETT (dcsjms@nus.edu.sg)
- Research topic: Group Testing.

MINDFUL Lab | University of Science – VNUHCM, Vietnam — Nov 2024 – Jun 2025
Research Assistant | Supervisor: Dr. Thach BUI (bvthach@fit.hcmus.edu.vn)
- Research topics: Group Testing, Matrix Completion.

AISIA Lab | Ho Chi Minh City, Vietnam — Mar 2024 – Jun 2025
Research Intern | Supervisor: Prof. Binh NGUYEN (ngtbinh@hcmus.edu.vn)
- Research topics: MM algorithms for Mixture-of-Experts architectures, Model selection for Mixture-of-Experts.

HONORS & AWARDS

Vietnam National Mathematics Olympiad — **First Prize** — 2024
Vietnam National Mathematics Olympiad — **Third Prize** — 2023
Vietnam Mathematics Summer School Contest — **First Prize** — 2022, 2023
Ho Chi Minh City Mathematics Olympiad — **First Prize** — 2024
30/4 Mathematics Olympiad — **First Prize** — 2023
Odon Vallet Scholarship — 2023, 2024
ASEAN Undergraduate Scholarship, NUS — AY2025/26

SKILLS & BACKGROUND

AI Foundations	Machine Learning, Deep Learning, Reinforcement Learning, Uncertainty Quantification.
Mathematics	Abstract Algebra, Linear Algebra, Real Analysis, Probability Theory, Statistics.
Programming	Python, C++, Java, JavaScript, MATLAB.
Languages	English (IELTS 7.5), Vietnamese (native), German (beginner).

RESEARCH

Research Interests. My primary research interest is **Uncertainty Quantification**, with previous work focusing on Conformal Prediction. I am currently exploring UQ for streaming and sequential processes, including LLM scheduling, reward hacking, and multi-agent debate.
Beyond my primary research focus, I have also contributed to projects in Group Testing, Optimal Transport, Mixture-of-Experts Optimization, and Computer Vision.

Uncertainty Quantification

1. **Conformal Prediction for Distribution-to-Distribution Regression** *(Accepted at NeurIPS 2026.)*
   Trung-Khang Tran[*], Tuan A. Hoang[*], Viet-Hoang Tran[*], Tan Minh Nguyen.
2. **Conformal Prediction under Distribution Shift: False Coverage Proportion Guarantees** *(Under submission.)*
   Trung-Khang Tran[*], Tuan A. Hoang[*], Viet-Hoang Tran[*], Tan Minh Nguyen.
3. **Split Conformal Prediction with Tunable Prediction Set Size** *(Under submission.)*
   Trung-Khang Tran[*], Tuan A. Hoang[*], Viet-Hoang Tran[*], Tan Minh Nguyen.

Other Research

1. **Tree-Sliced Orlicz Integral Probability Metric** *(Accepted at NeurIPS 2026.)*
   Trung-Khang Tran[*], Tuan A. Hoang[*], Viet-Hoang Tran[*], Tan Minh Nguyen.
2. **MELCOT: A Hybrid Learning Architecture with Marginal Preservation for Matrix-Valued Regression** *(Accepted at WSDM 2026.)*
   Khang Tran, Hieu Cao, Thinh Pham, Nghiem Diep, Tri Cao, Binh Nguyen.
3. **Revisiting Tree-Sliced Wasserstein Distance Through the Lens of the Fermat–Weber Problem** *(Accepted at ICLR 2026.)*
   Viet-Hoang Tran, Thanh Tran, Thanh Chu, **Trung-Khang Tran**, Duy-Tung Pham, Tam Le, Tan Minh Nguyen.
4. **Tree-sliced Sobolev IPM** *(Accepted at ICLR 2026.)*
   Viet-Hoang Tran, Thanh Tran, Thanh Chu, Duy-Tung Pham, **Trung-Khang Tran**, Tam Le, Tan Minh Nguyen.
5. **Group Testing with Selectable Thresholds** *(Under submission.)*
   Trung-Khang Tran, Daniel McMorrow, and Jonathan Scarlett.
6. **Matrix Completion in Group Testing: Bounds and Simulations** *(Under submission.)*
   Trung-Khang Tran and Thach V. Bui.
7. **Fast Model Selection and Stable Optimization for Softmax-Gated Multinomial-Logistic Mixture of Experts Models** *(Under submission.)*
   TrungKhang Tran[*], TrungTin Nguyen[*], Md Abul Bashar, Nhat Ho, Richi Nayak, Christopher Drovandi.
8. **Revisiting Incremental Stochastic Majorization-Minimization Algorithms with Applications to Mixture of Experts** *(Under submission.)*
   TrungKhang Tran[*], TrungTin Nguyen[*], Tung Doan, Binh T. Nguyen, Hien Duy Nguyen, Florence Forbes, Gersende Fort, Christopher Drovandi.
9. **High-dimensional Many-to-many-to-many Mediation Analysis** *(Under submission.)*
   TienDat Nguyen, **TrungKhang Tran**, CongKhanh Truong, Duy-Cat Can, Binh T. Nguyen, Oliver Y. Chén.
10. **Sparse Partial Optimal Transport via Quadratic Regularization** *(Accepted at Journal of Computer Science.)*
    Khang Tran[*], Khoa Nguyen[*], Anh Nguyen, Thong Huynh, Son Pham, Sy-Hoang Nguyen-Dang, Manh Pham, Bang Vo, Mai Ngoc Tran, Dung Luong.
11. **LGCA: Enhancing Semantic Representation via Progressive Expansion** *(Accepted at SoICT 2025.)*
    Cao Thanh Hieu, **Tran Trung Khang**, Pham Gia Thinh, Diep Tuong Nghiem, Nguyen Thanh Binh.
12. **CovHuSeg: An Enhanced Approach for Kidney Pathology Segmentation** *(Accepted at CSoNet 2025.)*
    Huy Trinh[*], **Khang Tran**[*], Nam Nguyen, Tri Cao, Binh Nguyen.